TASK SPECIFICATION: P6-02 PHASE P6 — ACADEMIC CAMPUS SERVICES & RESOURCE PODS

Tutoring Booking & Automated 70 SAR/Hour Escrow Transfer

Git Branch: `feature/P6-02-tutoring-booking` · Target Pull Request: feature/P6-02-tutoring-booking

Core Objective & Technical Scope:
Session enrollment engine executing atomic fee transfers (70 SAR/hr standard rate) from student wallet to tutor tuition credit account.

Priority Level	Critical	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P6-02-tutoring-booking`
Upstream Dependencies	P6-01, P4-02		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement enrollment service enforcing student capacity limits.
2. Execute atomic wallet transfer deducting 70 SAR/hr and crediting tutor ledger.
3. Prevent duplicate enrollment and tutor self-booking.
4. Expose booking endpoint with confirmation notifications.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `tutoring/application/*`
- `tutoring/api/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Insufficient wallet balance blocks booking with clear error message.
- Standardized 70 SAR/hr rate enforced strictly on backend.

VERIFICATION & TEST COVERAGE STRATEGY

- TutoringBookingServiceTest
- ConcurrentEnrollmentTest
- BookingControllerTest